

READ THIS FIRST — this document was written before real-genome validation

Module 3 has since been measured against **42 real bacterial genomes with 350 laboratory-measured susceptibility results**. On that test it scores **0.526 accuracy**, which with 176 resistant and 174 susceptible comparisons is approximately **chance**. No threshold changes this; the sweep peaks at 0.503.

Every accuracy figure in this document is measured on **constructed** isolates — generated background sequence with real resistance genes planted into it. Those figures are correct for constructed isolates and must not be presented as real-world performance.

What did transfer is **gene detection**: on real genomes it found genuine determinants (mecA in 9 real *S. aureus*, blaSHV in 5, plus blaNDM-1, blaVIM, blaCTX-M-15) at a 13.8% false-resistance rate, though it misses 73% of laboratory-confirmed resistance because 21 reference genes cannot cover real resistance. Full analysis: **reports/AUDIT_RESPONSE.md**.

TRIALSENSE · SEGUE 3.0 · TECHNICAL REPORT

Module 3 — Feature-by-Feature Technical Documentation

Tech stack, models, datasets, training, validation, accuracy, dependencies, limitations and future scope for each of the six features. Written for readers with an engineering background and no healthcare training.

FEATURES	ML MODELS	HELD-OUT SCORE	VALIDATION CHECKS	AUTOMATED TESTS
6	1 see §1.3	0.763 macro-F1	8	51
LINES OF CODE	12,084			

01 Introduction

1.1 · WHAT THE MODULE DOES

Module 3 takes the genetic sequence of one bacterial sample and determines which antibiotics that sample can already survive. It then answers five further questions about what that means for a company deciding whether to fund a drug development programme.

The input is a text file of A, C, G and T characters. The output is six separate reports. They are kept separate rather than combined into one score, because a single number would hide cases where the six answers disagree.

1.2 · SHARED TECHNICAL FOUNDATION

All six features run on the same base stack. Per-feature sections list only what is additional or specific.

COMPONENT	VERSION	ROLE
Python	3.13.5	Runtime
NumPy	2.5.3	Array maths, all sequence processing
scikit-learn	1.9.0	The classifier and its calibration
SciPy	1.18.1	Statistical helpers (via scikit-learn)
pandas	3.0.5	Tabular handling
Plotly	7.0.0	All charts
Streamlit	1.63.0	Web interface
RDKit	2026.03.6	Used by Modules 1–2, not Module 3

No paid services, no network calls at runtime, no GPU. The full system runs on a laptop CPU.

1.3 · WHAT IS, AND IS NOT, MACHINE LEARNING

READ THIS BEFORE DESCRIBING THE SYSTEM TO ANYONE

Module 3 contains **one** machine-learning model. The other components are similarity matching, lookup tables and statistics. Describing all six features as “AI” would be inaccurate, and the distinction is easy to defend once stated.

COMPONENT	CATEGORY	WHY BUILT THIS WAY
Resistance classifier	Machine learning	The only part where the correct answer must be inferred from data
Trend forecaster	Fitted statistical model	A curve fit to real figures, not a learned model
Gene search	Similarity matching	Comparison against known published sequences is exact and verifiable
Mechanism logic	Lookup table	Established published science; inferring it would be worse
Unknown-element scan	Statistical anomaly detection	Needs no training data, so it generalises to anything
Cost model	Arithmetic on user inputs	Must be auditable line by line

1.4 · THE TWO-LAYER ARCHITECTURE

Gene search and the classifier answer the same question by different means. Gene search is exact but only recognises the 21 genes we hold. The classifier covers anything but is less reliable.

Where gene search finds a definite match, its result takes priority. Measured effect: the classifier alone produces 5 missed detections across the 12 bundled samples; with the override, **zero**.

02 Feature 1 – Runway

Resistance projection to launch year

Estimates how resistant the target will be when the drug reaches market, not today.

TECH STACK	Python, NumPy (curve fitting via <code>numpy.polyfit</code> on transformed values), Plotly for the projection chart. Implemented in <code>trialsense/surveillance.py</code> , 1,215 lines.
MODEL	A logistic (S-shaped) curve, fitted by least squares after transforming the percentages onto a logit scale. This is a statistical fit, not a trained model. Why logistic and not a straight line: resistance cannot exceed 100%. Klebsiella carbapenem resistance rises about 2.5 percentage points a year; a straight line extended to 2035 produces a figure above 110%, which is not a valid number. A logistic curve flattens as it approaches a ceiling, which is also how the underlying process behaves. Why a transform instead of an optimiser: with nine annual data points a non-linear optimiser can fail to converge. The logit transform makes the problem linear, so the fit always returns a result.
DATASET	ICMR AMRSN Annual Reports, 2015–2024. India's national antimicrobial resistance surveillance network. 7 organisms, 46 organism-antibiotic series, 408 individual data points. Why chosen: it is the authoritative national source for the market this product targets, it is free and public, and it spans enough years to fit a trend. A second source, the WHO Global Health Observatory API, is included so the interface can show that independent surveillance programmes report different figures for the same organism and year.
TRAINING	No training in the machine-learning sense. The curve is fitted at runtime to whichever organism and antibiotic group the user selects. The 2015 figures are deliberately excluded from every fit. That year covered only blood and spinal-fluid samples, while 2016 onward covers all sample types. Combining them produces a false step in the data — the methicillin-resistance figure reads 42.6% for 2015 and 28.4% for 2016, and both are correct for their own sample sets.

VALIDITY

Time-split backtest. The curve is fitted using only data up to 2020, then asked to predict 2021–2024 — four years it has never seen. The predictions are compared against what actually happened.

This is the only component in Module 3 tested against real-world outcomes rather than labels we assigned.

ACCURACY

Predictions evaluated	184
Series covered	46
Mean absolute error	4.93 pp
Median absolute error	3.55 pp
Within 5 percentage points	64%
Within 10 percentage points	85%

In plain terms: when it predicts a resistance figure for a year it has not seen, it is typically within about five percentage points.

DEPENDENCIES

- Feature 6 (Spread) supplies a multiplier that shortens the projection when the detected gene is of a type that transfers rapidly.
- The core gene search supplies which genes are present.
- Independent of Modules 1 and 2.

LIMITATIONS

- It is an extrapolation. It is drawn as a dashed line inside a widening uncertainty band so it cannot be mistaken for measured data.
- Only the 7 organisms ICMR reports are covered. For others — including tuberculosis — the feature returns “no data available” rather than substituting a proxy figure.
- It assumes the trend continues. A new drug, a policy change or an outbreak would break that assumption, and nothing in the model would detect it.
- Surveillance sources disagree with one another by up to 15 percentage points for the same organism and year.

FUTURE SCOPE

- Add state-level rather than national figures, so projections reflect the region a trial would actually run in.
- Add confidence weighting by sample size — some series rest on a few hundred samples, others on tens of thousands.
- Extend beyond India using WHO country data, for companies targeting export markets.

HOW IT REDUCES COST

A company can spend nine years building a product for a market that has disappeared by the time it is finished. This feature identifies that before any money is committed. The saving is the entire development budget, not a portion of it.

03 Feature 2 — Rescue

Mechanism analysis and workaround assessment

Determines whether detected resistance can be worked around, or whether the programme must be abandoned.

TECH STACK

Pure Python. No model, no numerical libraries. Implemented in `trialsense/mechanisms.py` (476 lines) and `trialsense/perdrug.py` (341 lines).

MODEL

None. This is a structured lookup over annotations attached to each of the 21 genes: how the gene physically blocks the drug, whether a second compound can counteract that, and which individual drugs are affected.

Why no model: these relationships are established published science. A model asked to infer them would be less accurate than reading them, and would not be able to cite a source.

DATASET

- **CARD** (Comprehensive Antibiotic Resistance Database) ontology — individual drug relationships, 177 extracted rows.
- **NCBI AMRFinderPlus Reference Gene Catalog**, database version 2026-08-07.1 — per-drug subclass fields.

Both are stored in `trialsense/reference_sequences/gene_drug_mappings.csv` with a source identifier and resolvable URL on every row.

Why both: they disagree in places. Rather than silently choosing one, both are retained and the disagreement is reported.

TRAINING

No training. The annotations were written from published references and then cross-checked against the two databases above. Discrepancies found during that check are recorded in the code.

VALIDITY

- Verified against CARD and NCBI for all 21 genes.
- 9 automated tests assert the decision logic, including that two genes producing the same antibiotic-group result return opposite verdicts.

- A test asserts that a drug absent from a sparsely curated database is never reported as unaffected — absence of a record is not evidence.

ACCURACY

Not applicable in the statistical sense; this feature makes no predictions. The relevant measures are coverage and correctness of lookup:

Genes with mechanism annotations	21 / 21
Genes present in the per-drug database	19 / 21
Gene–class pairs resolvable to individual drugs	14
Documented database disagreements recorded	2

DEPENDENCIES

Requires the core gene search to identify which genes are present. Supplies its verdict to Feature 3, which uses it to distinguish a blocked candidate that could be recovered from one that could not.

LIMITATIONS

- Only works for the 21 genes we hold. An unrecognised gene produces no mechanism assessment.
- Three genes have only group-level entries in both databases, so no individual-drug detail is available for them.
- Two genes cause resistance through a single-character change in the DNA rather than through an added gene. Our method detects that the region is present but cannot determine whether the change is there.

FUTURE SCOPE

- Read gene annotations directly from CARD at build time instead of from a stored extract, so they update automatically.
- Add read-level variant detection to resolve the two single-character cases.
- Extend coverage beyond 21 genes; CARD holds several thousand.

HOW IT REDUCES COST

It works in both directions. It prevents a team abandoning a programme that a formulation change would have saved, and prevents a team spending months attempting a fix that cannot work. Two genes can produce an identical result on screen and require opposite decisions; without this feature that distinction is invisible.

04 Feature 3 – Portfolio

Multi-candidate, multi-target screening grid

Tests every candidate drug against every target organism and ranks them.

TECH STACK	Python, Plotly heatmap, Streamlit caching to avoid re-analysing the same organism once per candidate. Implemented in <code>trialsense/portfolio.py</code> , 341 lines.	
MODEL	None of its own. It orchestrates repeated calls to the core classifier and to Feature 2, then ranks the results. Ranking is by number of organisms the candidate remains viable against, with mean predicted resistance as a tie-break.	
DATASET	Inherits from the core classifier. For validation it additionally uses the ICMR national figures described under Feature 1.	
TRAINING	None. Every result is computed at request time.	
VALIDITY	Cross-check against national data. Because this feature produces no data of its own, it would otherwise have no validation claim. Every prediction in the grid is compared against the real ICMR figure for the same organism and antibiotic group. The two numbers measure different things — ours describes one sample, ICMR describes a national population — so each comparison is judged against its expected direction rather than treated as a target to match.	
ACCURACY	Comparisons with an ICMR reference	49
	Track the national figure within 15 pp	14
	Differ in the expected direction	34
	Flagged for review	16
	Mean gap across aligned comparisons	5.1 pp

Separately, 2 automated tests assert the grid is constructed correctly and that an unscreened antibiotic group is marked as not assessed rather than scored zero.

DEPENDENCIES

- Core classifier — supplies every cell value.
- Feature 2 — supplies the recoverable/not-recoverable distinction.
- Feature 1's dataset — supplies the national figures for cross-checking.

This is the most dependent feature in the module. If the classifier is wrong, the grid is wrong.

LIMITATIONS

- Inherits every limitation of the classifier.
- The ranking rule is a design decision, not a validated one. No evidence establishes that breadth of viability is the correct primary criterion.
- Cost, manufacturing difficulty and regulatory status are not considered — only resistance.

FUTURE SCOPE

- Allow the user to weight the ranking criteria.
- Accept multiple uploaded sample files rather than only bundled organisms.
- Export the grid in a format a portfolio committee can annotate.

HOW IT REDUCES COST

Producing the same comparison by laboratory testing takes weeks and costs lakhs. Here it takes seconds. It also changes the question from “is this candidate viable” to “which of ours is most viable”, which is the decision that actually allocates the budget.

05 Feature 4 – Unknown

Unidentified acquired-sequence detection

Reports DNA that appears to have come from another organism but matches nothing known.

TECH STACK	Python, NumPy. Implemented in <code>trialsense/novelty.py</code> , 504 lines.	
MODEL	Statistical anomaly detection, not a trained model. The sequence is divided into overlapping 900-character windows stepped 150 characters apart. For each window the frequency of every 4-character combination is measured, giving 256 numbers. Windows already explained by an identified gene are removed. A reference profile is estimated from what remains, using an iteratively trimmed average so that a few unusual windows cannot distort it. Any remaining window that differs sufficiently from that reference is reported. A window must exceed two separate thresholds — a relative one (8 robust deviations) and an absolute one (0.18). The relative measure alone becomes unstable when few windows remain.	
DATASET	None. This feature requires no training data, which is why it can operate on organisms the system has never seen.	
TRAINING	None. Thresholds were calibrated by measurement, described below.	
VALIDITY	Tested in two directions. Specificity: the 12 unmodified bundled samples are scanned; every gene in them is known, so any report is a false alarm. Sensitivity: synthetic sequences matching no reference are inserted into 7 samples and the scan must find them.	
ACCURACY	False alarms on unmodified samples	0 / 12
	Inserted unknown elements detected	7 / 7
	Highest score on a clean sample	0.118
	Lowest score on a sample with an insert	0.232

The detection threshold sits between those two values with roughly a twofold margin on each side.

DEPENDENCIES	Requires the core gene search to know which regions are already explained. The model-distribution indicator additionally requires the trained classifier, which stores a summary of its training data.
LIMITATIONS	• Tested only against inserts we created. It has never encountered a genuinely novel resistance gene, because we have no example of one. • It reports that something unidentified is present. It does not identify it and makes no claim that it causes resistance. • Requires at least 8 windows. Short sequences return “not assessed” rather than a clean result. • If most of a sequence is already explained by known genes, too little remains to estimate a reference profile, and the scan declines to run.
FUTURE SCOPE	• Validate against genes deliberately withheld from the reference set, as a closer proxy for genuine novelty. • Report which known gene family an unidentified region most resembles. • Feed flagged regions directly into a public database search.
HOW IT REDUCES COST	Every comparable tool reports only what is in its database, so silence from such a tool is indistinguishable from confirmation that nothing is present. Those are very different statements. This feature prevents a company proceeding on a clearance that was never actually given.

06 Feature 5 – Value

Cost-of-late-discovery calculator

Converts a finding into the spending that would have been committed before it surfaced otherwise.

TECH STACK	Python, Plotly waterfall chart, Streamlit sliders. Implemented in <code>trialsense/economics.py</code> , 272 lines.
MODEL	Arithmetic over user-supplied inputs. It totals the cost of each development stage up to the point the problem would normally surface, then multiplies by the probability the programme survives long enough to reach that point. The framework is the expected value of information, a standard decision-analysis approach. The calculation is deliberately simple enough to verify by hand from the figures shown.
DATASET	STATE THIS PLAINLY WHENEVER THE FEATURE IS SHOWN There is no dataset. The default stage costs and success probabilities are order-of-magnitude figures supplied as starting values, not measurements. Every one is an adjustable slider, and the interface says so on the same screen.
TRAINING	None.
VALIDITY	Two automated tests. One asserts that a triggered finding produces a positive figure with a probability weighting applied. The other asserts that a clean result produces zero — the feature must never present the absence of a problem as a saving. No external validation is possible or claimed. No drug programme has been run through this system.
ACCURACY	Not applicable. This feature measures nothing; it computes a consequence from assumptions the user supplies. Its correctness is arithmetic correctness only.

DEPENDENCIES

Requires a resistance result from the core scan to determine whether a finding was triggered at all.

LIMITATIONS

- Entirely dependent on inputs the user provides. Poor inputs produce a poor figure, and the feature cannot detect this.
- Assumes the problem would have surfaced at one identifiable stage. In practice it might surface earlier, later, or not at all.
- Ignores the value of information gained along the way, and any parallel programme that might have absorbed the work.

FUTURE SCOPE

- Preload published industry cost figures with citations, so defaults are sourced rather than estimated.
- Add a sensitivity view showing which input most affects the result.
- Allow a company to save its own cost profile.

HOW IT REDUCES COST

It does not reduce cost directly. It **measures** the reduction the other features produce, in the format required by the person who authorises spending. A scientific finding that cannot be expressed financially does not reach that person.

07 Feature 6 — Spread

Transfer-rate assessment

Reports how quickly a detected resistance gene moves through a bacterial population.

TECH STACK	Pure Python, Plotly bar chart. Implemented in <code>trialsense/mechanisms.py</code> alongside Feature 2.
MODEL	A lookup. Each gene is annotated with the type of genetic element it sits on. Some elements can be passed directly between bacteria, including between different species; others are fixed in the main chromosome and pass only to descendants. Each type carries a multiplier, from 1.0 for fixed genes to 2.5 for freely transferable ones. The fastest-moving gene present determines the multiplier for the sample, which is then applied to Feature 1's projection.
DATASET	The element-type classifications come from published references and were verified against NCBI and CARD annotations for all 21 genes. The multipliers themselves are not from a dataset — see Limitations.
TRAINING	None.
VALIDITY	3 automated tests assert that a transferable gene scores higher than a fixed one, that the fastest gene present governs the result, and that a sample with no detected genes returns a neutral multiplier rather than an error.
ACCURACY	The classifications are verifiable and were checked for all 21 genes. The multipliers are not measured quantities and no accuracy figure is claimed for them.
DEPENDENCIES	Requires the core gene search. Feeds its multiplier into Feature 1.

LIMITATIONS

PRINCIPAL LIMITATION

The multipliers are reasoned estimates of relative magnitude, not values fitted to data. They are defensible as ordering — transferable genes do spread faster than fixed ones — but the specific numbers are not measured. The interface states this.

FUTURE SCOPE

- Derive the multipliers from observed historical spread rates rather than assigning them.
- Account for the fact that transfer rates differ by organism and setting.
- Detect the transferable element directly in the sequence rather than inferring it from the gene's identity.

HOW IT REDUCES COST

It corrects the planning timeline. A team working to an unadjusted projection is planning against a deadline further away than the real one, which affects how much development delay the programme can absorb.

08 The core classifier

Every feature above depends on this component, so it is documented separately. It is the only machine-learning model in Module 3.

Multi-label resistance classifier

Estimates resistance across 12 antibiotic groups from a genetic sequence.

MODEL

```
MultiOutputClassifier          12 labels, one per antibiotic group
└─ CalibratedClassifierCV      method=isotonic, cv=3
    └─ RandomForestClassifier  n_estimators=250
                                min_samples_leaf=2
                                max_features=sqrt
                                random_state=42
```

Why a random forest: it performs well on small datasets, needs no GPU, and trains in under two minutes on a laptop.

Why calibration was added: the uncompensated forest was systematically under-confident — when it reported 55%, the observed rate was 97%. Because that number is multiplied into a downstream score, the understatement propagated. Isotonic regression corrects it.

INPUT REPRESENTATION

Each sequence is converted into the frequency of every 5-character combination, producing **1,024 numbers**. This is a standard alignment-free representation and scales linearly with sequence length, so it works on a complete genome as readily as on a fragment.

DATASET

700 constructed samples. Each is assembled from real published gene sequences placed into generated species-typical background DNA.

Real: all 21 gene sequences, from the NCBI Bacterial Antimicrobial Resistance Reference Gene Database, version 2026-08-07.1 — the same curated set NCBI's own tool uses. Accession numbers appear in every file header.

Constructed: the background DNA, and the choice of which genes each sample carries. Complete annotated genomes are substantially larger than a demonstration can distribute. Gene combinations are sampled to reflect which genes genuinely occur together.

TRAINING

`python train.py`, approximately two minutes on a laptop CPU. It generates the 700 samples, converts each to its 1,024-number representation, fits the calibrated forest, records a summary of the training distribution for later out-of-distribution checks, and writes the evaluation figures.

Fully deterministic — fixed random seeds throughout, so the same command reproduces the same model.

ACCURACY

EVALUATION	MACRO-F1	PER-LABEL	EXACT MATCH
Random split	0.900	95.1%	58.9%
Unseen gene combinations	0.763	94.3%	51.1%
Entire species withheld	0.897	—	—

Quote 0.526, the real-genome figure. The 0.763 above is the hardest CONSTRUCTED test — the model is evaluated on gene combinations absent from its training data — but constructed is the operative word. On 42 real genomes with laboratory AST the model scores 0.526, approximately chance, and that is the number that describes real-world behaviour.

ERROR TYPES, REPORTED SEPARATELY

Missed detections (reporting susceptible when resistant) end a programme; false alarms (the reverse) cost one confirmatory test. The two are counted separately because the costs are not comparable.

	MISSED	FALSE ALARM	CORRECT
Classifier alone	3.15%	14.39%	—
Deployed, across 144 calls	0	0	100%

The deployed figure is an integration check on samples whose contents we chose, not a held-out test. **0.526 on real genomes is the figure to quote.**

LIMITATIONS

- Trained on constructed sample scaffolds with labels we assigned, not laboratory-confirmed results.
- On its own the missed-detection rate exceeds the standard applied to commercial testing equipment. It is not deployed on its own.
- Real sequencing data carries noise, contamination and incomplete assembly that our samples do not.

FUTURE SCOPE

- **Highest priority:** retrain on complete genomes from NCBI Pathogen Detection, which carry laboratory-confirmed results. This addresses both remaining data limitations at once.
- DNABERT-2, a genomic language model, is integrated as an optional comparison path with a ready-to-run notebook for free cloud GPU. It has not been run. We expect it to underperform on constructed scaffolds, and consider measuring that more useful than asserting it.

09 Defects found and corrected

Recorded because a validation process that never finds anything is not evidence of correctness. Each of these was found by our own testing.

DEFECT	EFFECT	RESOLUTION
Unscreened antibiotic group reported as 0%	A drug from a group we do not check was reported as facing no resistance, rather than as unchecked	Now reports “not screened”, and is excluded from the combined score
Incorrect position number on one gene	A resistance marker was labelled with the position used for a different organism	Corrected throughout after verification against the reference sequence
Three India-relevant genes absent	Coverage of genes reported in Indian surveillance was 5 of 8	Added; coverage now 8 of 8
Unknown-element scan re-detecting known genes	Unmodified samples scored higher than deliberately modified ones	Known regions are now masked before scoring; rebuilt and recalibrated
Detection threshold calibrated on constructed sequences	After switching to real sequences the gene search matched almost everything	Recalibrated from 0.55 to 0.83 by measurement
Model under-confident in its mid-range	Reported percentages were systematically below observed rates	Isotonic calibration added; error reduced from 0.100 to 0.015
Clean control sample reported as elevated risk	A single borderline group pushed a sample with no detected genes above threshold	Separate screening and confirmation thresholds introduced
Incorrect “unaffected drug” claim	A drug absent from a sparsely curated database was reported as unaffected	Absence of a record is no longer treated as evidence; test added

10 Reproducibility and provenance

10.1 · REGENERATING EVERY FIGURE IN THIS REPORT

```
conda deactivate          # if a conda environment is active
.venv\Scripts\activate    # prompt must show (.venv)
pip install -r requirements.txt

python train.py            # ~2 min  -> models/, metrics.json
python validate_module3.py # ~2 min  -> models/validation.json
python smoke_test.py       # ~2 min  -> 51 checks
streamlit run app.py
```

All figures in this report are read from `models/metrics.json` and `models/validation.json`. Random seeds are fixed, so the commands above reproduce the same values.

10.2 · DATA PROVENANCE

DATA	SOURCE	USED BY
21 gene sequences	NCBI AMRFinderPlus reference database 2026-08-07.1	All features
Resistance figures 2015–2024	ICMR AMRSN Annual Reports	Features 1, 3
National figures, cross-check	WHO Global Health Observatory API	Feature 1
Per-drug relationships	CARD ontology; NCBI Reference Gene Catalog	Feature 2
Gene prevalence in India	ICMR AMRSN 2018 molecular chapter	Validation
Stage costs, success rates	None — user-supplied assumptions	Feature 5

10.3 · TERMS USED IN THIS REPORT

Resistance	A bacterium's ability to survive an antibiotic that would normally kill it
Gene	A section of DNA. Certain genes cause resistance
Antibiotic group	A set of antibiotics that work in a similar way. We track 12
Sample / isolate	One bacterial specimen
Macro-F1	An accuracy measure from 0 to 1 that treats all 12 groups equally regardless of how often each occurs
Held-out test	Evaluation on data deliberately excluded from training
Calibration	Whether a stated probability matches the observed rate
Percentage point (pp)	The difference between two percentages. 66% to 84% is 18 pp

11 Conclusion

WHAT HAS BEEN BUILT

Six features over one machine-learning model and a set of verifiable reference data. The system runs on a laptop with no paid services and no network access at runtime. It is covered by 51 automated tests and 8 independent validation checks.

WHAT THE MEASUREMENTS ESTABLISH

CLASSIFIER 0.763 unseen gene combinations	FORECASTER 4.93 pp 184 unseen predictions	GENE SEARCH 12 / 12 no false matches	UNKNOWN SCAN 0 / 7 false alarms / detections missed	DEPLOYED ERRORS 0 across 144 calls
GENE COVERAGE 8 / 8 India-relevant determinants				

THE PRINCIPAL LIMITATION

STATED ONCE, CLEARLY

The gene sequences are real; the sample scaffolds around them are constructed, and the gene contents are assigned by us rather than laboratory-confirmed. Every accuracy figure in this report should be read with that in mind.

Replacing the constructed scaffolds with complete genomes from NCBI Pathogen Detection would resolve both points at once, and is the single highest-value improvement available.

A MEASUREMENT WORTH RECORDING

The gene sequences were originally constructed rather than real. Replacing them with published sequences **reduced** measured accuracy from 0.99 to 0.92, and to 0.763 on the hardest test. The change was retained, because the reduction represents the degree to which constructed data had been overstating performance.

The deployed system nonetheless improved, because gene search on real sequences is exact and now takes priority over the model where it applies.

TrialSense is a research screening prototype for pharmaceutical research and development. It is not a diagnostic tool and not a clinical decision-support system. No output should be used to treat anyone.

All figures generated from `models/metrics.json` and `models/validation.json` .
Reproduce with the commands in §10.1.